

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
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I want to work on the instruments Europe is actually using to decarbonise, and to do it inside a European research group. That is why I specialised in energy economics at Cologne, and why I chose the EU Emissions Trading System for my final seminar paper rather than a topic closer to the labour economics my thesis came out of. What draws me to auction design is that the interesting variation sits in the rules. A support scheme is a set of commitments, deadlines and penalties, and what a firm will build depends on those more than on any headline price. I would like to spend four years establishing which of those rules do the work.

My training is Cologne through the Institute of Energy Economics. I took *Energy Markets and Regulation* with Jun.-Prof. Dr. Oliver Ruhnau, whose chair is in energy market design, and finished it with 1.3 on a scale where 1.0 is the maximum, alongside econometrics at 1.3 and mathematics at 1.0. My master's thesis was graded 1.0. It estimates the labour-market effects of a national scholarship programme in Brazil on a panel I built from administrative records, roughly five million linked employer-employee observations across 558 microregions from 2005 to 2019, in R through BigQuery. Using Callaway and Sant'Anna (2021) with doubly-robust inference, the programme raises wages by 3.04 per cent in low-exposure regions, distinguishable from zero at the 10 per cent level, along a dose-response that is non-monotonic in exposure. In my seminar paper I define the effective carbon cost as the allowance price attenuated by the sector-year free-allocation share, and defend that against the lump-sum objection using the activity thresholds and closure provisions that tie EU allocation to continued production.

Of the four directions in the call I would take up pre-qualification with completion bonds, and local content requirements, because both fix what a bid means before it is awarded. I would want to measure them as well as model them. Member states introduced auctions on different dates and have since revised penalties, bid bonds, realisation deadlines and technology lots on different national timetables, which is the variation my thesis was built to handle. Three colleagues at CORE make this the place to do it. Your work with Yueting Yu on discriminatory against uniform-price auctions asks how short-run bidding maps into long-run investment, and a penalty regime that changes what a bid commits to changes that map. Quentin Lété works on market design and the efficiency cost of zonal pricing with market-based re-dispatch, the normative counterpart to a decade of European practice. Enerpol, with Jean Hindriks and François Maniquet, studies who bears the cost of the transition, and distributional estimation across regions and groups is what my thesis does.

In ten years I want to be producing the evidence European energy policy is made on, whether from a research group of this kind or from inside a European institution. I have been preparing for the second of those in parallel, and I would rather arrive there as someone who has done the research than as someone who has only read it. This position is the right first step for both. It is supervised by the person working on the auction-format question I want to answer, it sits in a group that runs theory, operations research and empirical work together, and it starts in autumn 2026. I would be glad to discuss the project with you.

Available from October 2026. French B1, German and Tunisian Arabic native, English IELTS Academic 8.0. EU citizen. Referees: Prof. Dr. Pia Pinger and Jun.-Prof. Dr. Oliver Ruhnau, contact details on the CV.

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